

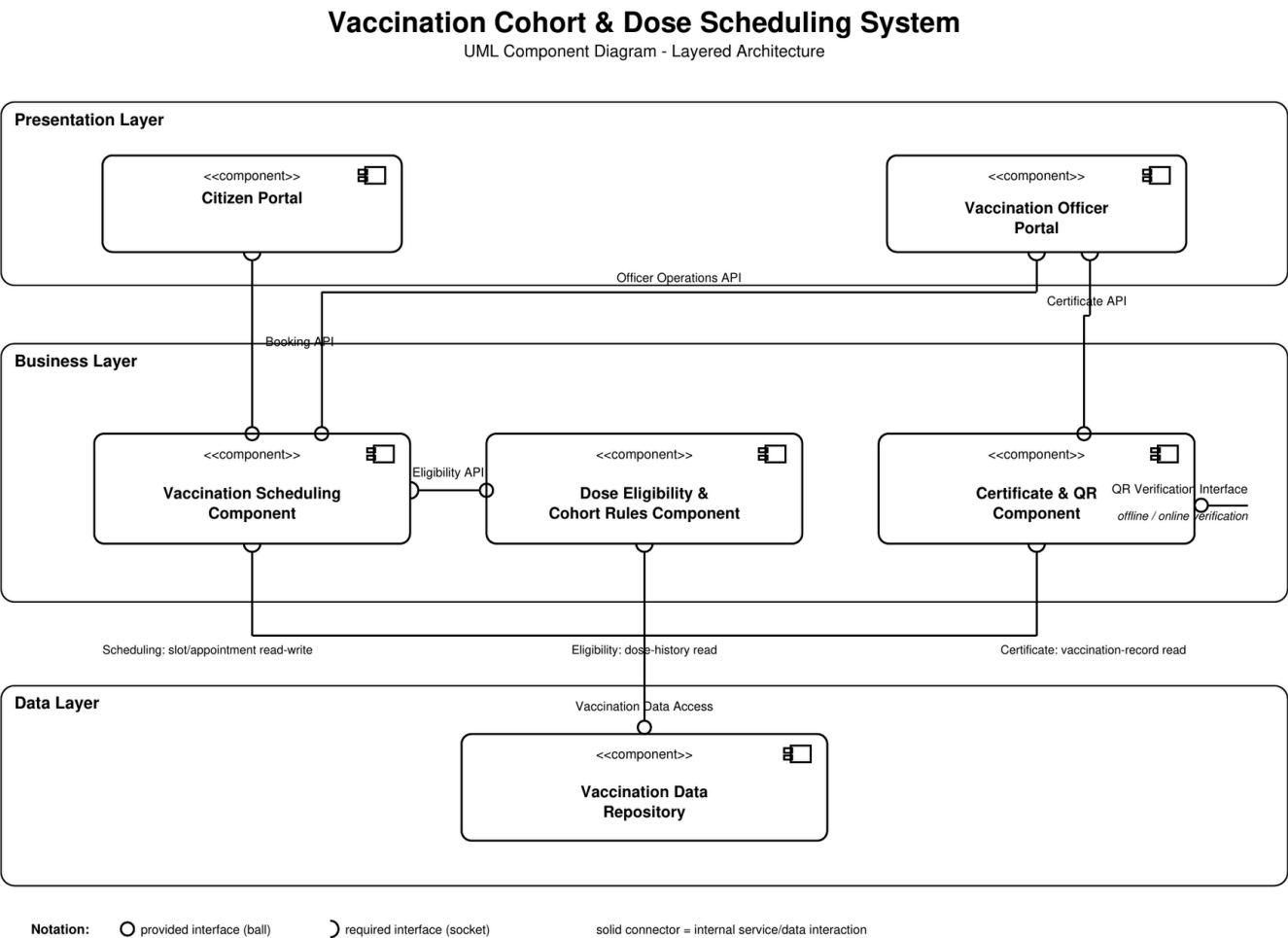
Lab-3 Architectural Pattern Selection

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#Vaccination Component Diagram



#My Architecture Justification

Architecture Selection:

We chose a Layered Architecture for the Vaccination Cohort & Dose Scheduling System.

Architectural Choice

The system is divided into Presentation, Business, and Data layers. Citizen and Vaccination Officer portals form the Presentation layer. Scheduling, dose-eligibility, and certificate/QR functions form the Business layer, while vaccination records and slot data are managed through the Data layer.

Reason 1 -

Clear separation of vaccination rules and user interfaces

Dose-interval validation is a core rule of the system. Keeping eligibility rules in a dedicated business component prevents booking screens or officer screens from bypassing the minimum-dose-interval logic. The same rule can therefore be reused consistently for citizen booking and officer operations.

Reason 2 -

Maintainability and controlled data access

Scheduling, certificate generation, and vaccination-record storage have different responsibilities. A layered design isolates these responsibilities, so changes to slot-allocation logic or QR-certificate generation can be made without redesigning the entire application. Database access is also centralized through the data layer instead of being spread across user-facing components.

Security Advantage

The Presentation layer does not access vaccination records directly. Requests pass through business services, where authentication, authorization, dose-rule checks, and certificate-signing controls can be applied before data is read or modified. This reduces unauthorized direct access to sensitive vaccination information and supports controlled QR-certificate verification.

Performance Benefit

Frequently used operations such as dose-eligibility checks and QR verification can be optimized inside their business components without changing the

portals or database structure. The architecture also avoids the additional inter-service network overhead that a microservices design would introduce for this scope, making low-latency verification easier to achieve.

Conclusion: Layered Architecture is the most suitable choice because it provides clear responsibility boundaries, centralized rule enforcement, controlled data access, and straightforward performance optimization while remaining simpler to develop and maintain than a distributed microservices solution.

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